

Project Plan and Research Questions

Context-Aware False Alarm Reduction for Violence Detection in Surveillance Videos Using a Lightweight Gating Module

Group 2 - DAPM391 - SU26 - FPT University

1. Problem Statement

The objective is to build a practical CCTV violence detection system that reduces False Positive Rate (FPR) without increasing False Negative Rate (FNR). The project combines an X3D-S base detector with a lightweight Context Gating Module (CGM) that uses crowd, lighting, and motion context features.

2. Proposed Pipeline

Tier 1: X3D-S estimates the base violence probability p_{base} from RGB video clips.

Tier 2: Annotation-free context streams extract crowd density, lighting quality, and motion-chaos features.

Tier 3: CGM combines p_{base} with 12 context features and outputs a calibrated final probability.

3. Research Questions

RQ	Question	Primary Evidence
RQ1	Does X3D-S + CGM reduce FPR compared with X3D-S alone?	E0 vs E4 ablation FPR
RQ2	Does FPR reduction create an unacceptable FNR trade-off?	E0 vs E4 ablation FNR
RQ3	Which context stream contributes most to the final prediction?	E1/E2/E3/E4 ablation
RQ4	Is motion synchrony redundant when crowd and lighting features are present?	E5 ablation without motion

4. Four-Week Timeline

Week	Focus	Deliverables
1	Literature and dataset validation	Related work, dataset protocol, approved RQs
2	Baseline training	Reproducible split, DataLoader test, X3D-S E0 baseline
3	Context extraction and CGM	p_{base} , crowd, lighting, motion cache, E1-E5 ablation
4	Validation and reporting	Final comparison table, figures, report, slides

5. Member Responsibilities

Ha Viet Hung: project planning, research questions, technical phases, motion-chaos stream, X3D-S fine-tuning, and CGM setup.

Nguyen Viet Nhan: RWF-2000/RLVS download and verification, data visualization, crowd features with YOLOv8n, lighting features with OpenCV, and comparison models.

Nguyen Thai Kiet: related paper summaries, advanced visualizations, missing-value and outlier handling, encoding with scikit-learn, and final metric comparison.

Tran Bao Nguyen: shared environment setup, analytical insight extraction, SQL/data support, report writing, and slide preparation.

6. Environment Summary

Python 3.10, PyTorch 2.x, CUDA, PyTorchVideo, Ultralytics, OpenCV, scikit-learn, pandas, and wandb.